

MOHAMMAD AFAQ

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mafaq

mafaq318

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Oulu, Finland

SUMMARY

Software Engineer II specializing in **C++14/C++20, Adaptive AUTOSAR, embedded Linux, and SDK/platform engineering**. Experienced in production software development, secure diagnostics, cross-compilation, target debugging, software design ownership, and requirements engineering across Linux, Windows/WSL, QEMU, and embedded ARM environments.

EXPERIENCE

Elektrobit Automotive Finland Oy

Software Engineer II

Oct 2022 – Present

Oulu, Finland

Software Engineer I, Oct 2022 – 2024

- Hold **design ownership of the Windows development SDK**, covering architecture, requirements, SwAD/SwDD documentation, integration, and maintenance.
- Led the SDK transition from **VirtualBox to WSL**, improving development environment performance by approximately **60%** while reducing reliance on Windows-native third-party OSS packages.
- Own and develop production **C++14/C++20 Adaptive AUTOSAR components** for a Linux-based automotive platform across design, implementation, integration, debugging, and maintenance.
- Develop **Secure Diagnostics** functionality for calibration read/write, ECU configuration access, and secure ECU lock/unlock operations.
- Work with **ara::com, SOME/IP, ARXML, manifests, CryptoAPI, and Botan**, applying event-driven processing, queues, templates, and smart-pointer-based ownership.
- Develop and maintain functionality for **Identity and Access Management (IAM)** and **eMMC usage monitoring** for Adaptive applications.
- Maintain SDK infrastructure involving **Linux/WSL, systemd, and cross-compilation toolchains**; enabled ARM target debugging from x86 development environments and resolved QEMU/Windows networking issues.
- Manage requirements, dependency integration, and verification across the SDK and multiple components using **gcov, Bullseye, Robot Framework, GoogleTest, and GoogleMock**.
- Provide **on-site customer support**, including travel to customer sites for EV truck project development, diagnostics, log collection, and debugging using **CAN, UART, and DLT**.

Teradata Global Consulting

Associate Consultant

Sept 2021 – Aug 2022

Islamabad, Pakistan

- Supported enterprise **ETL/DCM systems** for retail-sector clients.
- Managed **SQL database operations** including performance tuning, monitoring, backup, and recovery, with supporting workflows in **Microsoft Azure**.
- Automated operational and service workflows to reduce manual effort and improve support efficiency.

Educative Inc

Technical Engineer

June 2021 – Sep 2021

Labore, Pakistan

- Worked with **Docker, Kubernetes, and Linux-based development environments**.
- Used **Shell scripting** and supported automated testing workflows, including Elixir-based testing.

EDUCATION

M.Sc. (Technology) in Computing Sciences
TAMPERE UNIVERSITY
Major: Signal Processing and Machine Learning

Aug 2022 – Mar 2026
Tampere, Finland

B.Sc. in Electrical Engineering
LAHORE UNIVERSITY OF MANAGEMENT SCIENCES

Sep 2017 – May 2021
Lahore, Pakistan

TECHNICAL STRENGTHS

Programming	C, C++14/17/20, Python, SQL, Shell Scripting
C++ Development	STL, Templates, Smart Pointers, Event-Driven Processing, Queues, Resource Ownership, Asynchronous Operations
AI-Assisted Development	LLMs, AI Agents, Coding Assistants, Prompt Engineering, Code Review, Debugging, Refactoring, Test Generation
Embedded & Automotive	Adaptive AUTOSAR, ar::com, SOME/IP, ARXML, Diagnostics, IAM, EmMC Statistics
Linux	Embedded Linux, systemd, Process & Resource Management, IPC, QEMU, Cross-Compilation, Toolchains, Target Debugging
Build & Tooling	CMake, GCC, Makefiles, Git, Yocto, Jenkins, CI/CD, Docker
Testing & Debugging	GDB, GoogleTest, GoogleMock, Robot Framework, gcov, Bullseye
Engineering	Software Design, SwAD, SwDD, Requirements, Code Reviews, Agile

RESEARCH & PROJECTS

Master's Thesis

Model Quantization for Efficient Differential Mobility Spectrometry (DMS) Analysis

- Evaluated multiple attention mechanisms in a **U-Net-based architecture** for DMS plot prediction.
- Investigated **model quantization** and analyzed trade-offs between prediction accuracy, memory footprint, and inference performance for embedded deployment.

Bachelor's Thesis

Detection and Removal of Motion Artifacts from Photoplethysmogram {PPG} Data

- Processed and analyzed PPG sensor data affected by motion and perfusion artifacts.
- Implemented real-time overlapping window based algorithm for artifact removal and improved heart rate accuracy.

CERTIFICATIONS

- CMake for Cross-Platform C++ Project Building
- C++ Unit Testing: Google Test and Google Mock
- Microsoft Azure Fundamentals Certified
- Teradata Vantage Associate Certified

HOBBIES & INTERESTS

- **Home Lab & Self-Hosting** — Building and maintaining home servers, experimenting with Linux services, networking, containers, and self-hosted applications.
- **Mobile Development** — Exploring mobile application development and building small personal projects.
- **Tennis**
- **Reading**